

Aptamer-functionalized lipid nanoparticles targeting osteoblasts as a novel RNA interference-based bone anabolic strategy

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Evidence abstract

Currently, major concerns about the safety and efficacy of RNA interference (RNAi)-based bone anabolic strategies still exist because of the lack of direct osteoblast-specific delivery systems for osteogenic siRNAs.

Here we screened the aptamer CH6 by cell-SELEX, specifically targeting both rat and human osteoblasts, and then we developed CH6 aptamer-functionalized lipid nanoparticles (LNPs) encapsulating osteogenic pleckstrin homology domain-containing family O member 1 (Plekho1) siRNA (CH6-LNPs-siRNA).

Our results showed that CH6 facilitated in vitro osteoblast-selective uptake of Plekho1 siRNA, mainly via macropinocytosis, and boosted in vivo osteoblast-specific Plekho1 gene silencing, which promoted bone formation, improved bone microarchitecture, increased bone mass and enhanced mechanical properties in both osteopenic and healthy rodents.

These results indicate that osteoblast-specific aptamer-functionalized LNPs could act as a new RNAi-based bone anabolic strategy, advancing the targeted delivery selectivity of osteogenic siRNAs from the tissue level to the cellular level.

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Evidence checklist

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Source continuity

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